

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

1	Medium Integrals	2
1.1	10 July 2026	2
1.2	9 July 2026	2
1.3	8 July 2026	3
1.4	7 July 2026	4
1.5	6 July 2026	4
2	Hard Integrals	6
2.1	10 July 2026	6
2.2	9 July 2026	6
2.3	8 July 2026	7
2.4	7 July 2026	8
2.5	6 July 2026	9

Medium Integrals

1.1 10 July 2026

Medium Integral

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\cos(x)}}{\left(\sqrt[3]{\sin(x)} + \sqrt[3]{\cos(x)}\right)^7} dx$$

Solution: We utilize the symmetry property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. Applying the transformation $x \rightarrow \frac{\pi}{2} - x$, the integral becomes:

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\sin(x)}}{\left(\sqrt[3]{\cos(x)} + \sqrt[3]{\sin(x)}\right)^7} dx$$

Adding the original integral and the transformed integral yields:

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\cos(x)} + \sqrt[3]{\sin(x)}}{\left(\sqrt[3]{\sin(x)} + \sqrt[3]{\cos(x)}\right)^7} dx = \int_0^{\frac{\pi}{2}} \frac{1}{\left(\sqrt[3]{\sin(x)} + \sqrt[3]{\cos(x)}\right)^6} dx$$

Divide the numerator and denominator by $\cos^2(x)$:

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{\left(\sqrt[3]{\tan(x)} + 1\right)^6} dx$$

Let $u = \tan(x)$, so $du = \sec^2(x) dx$. The limits change from $0 \rightarrow \infty$:

$$2I = \int_0^{\infty} \frac{1}{(u^{1/3} + 1)^6} du$$

Next, substitute $v = u^{1/3} + 1$. This implies $u = (v-1)^3$ and $du = 3(v-1)^2 dv$. The limits are now $1 \rightarrow \infty$:

$$2I = \int_1^{\infty} \frac{3(v-1)^2}{v^6} dv = 3 \int_1^{\infty} \frac{v^2 - 2v + 1}{v^6} dv = 3 \int_1^{\infty} (v^{-4} - 2v^{-5} + v^{-6}) dv$$

Integrating term by term:

$$2I = 3 \left[-\frac{1}{3}v^{-3} + \frac{1}{2}v^{-4} - \frac{1}{5}v^{-5} \right]_1^{\infty} = 3 \left(0 - \left(-\frac{1}{3} + \frac{1}{2} - \frac{1}{5} \right) \right) = 3 \left(\frac{10 - 15 + 6}{30} \right) = 3 \left(\frac{1}{30} \right) = \frac{1}{10}$$

Thus, the final value of the integral is:

$$I = \frac{1}{20}$$

1.2 9 July 2026

Medium Integral

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin(\theta) \sin(2\theta) (\sin^6(\theta) + \sin^4(\theta) + \sin^2(\theta)) \sqrt{2 \sin^4(\theta) + 3 \sin^2(\theta) + 6}}{1 - \cos(2\theta)} d\theta$$

Solution: First, simplify the integrand using double-angle identities: $1 - \cos(2\theta) = 2 \sin^2(\theta)$ and $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$. The numerator contains the product $\sin(\theta) \sin(2\theta) = 2 \sin^2(\theta) \cos(\theta)$. The $2 \sin^2(\theta)$ perfectly cancels out the denominator, leaving:

$$I = \int_0^{\frac{\pi}{2}} \cos(\theta) (\sin^6(\theta) + \sin^4(\theta) + \sin^2(\theta)) \sqrt{2 \sin^4(\theta) + 3 \sin^2(\theta) + 6} d\theta$$

Let $x = \sin(\theta)$, $dx = \cos(\theta) d\theta$. The limits become 0 to 1:

$$I = \int_0^1 (x^6 + x^4 + x^2) \sqrt{2x^4 + 3x^2 + 6} dx$$

Factor out x^2 from the polynomial to get $x^2(x^4 + x^2 + 1)$. We can push one factor of x into the square root (which becomes x^2 inside):

$$I = \int_0^1 x (x^4 + x^2 + 1) \sqrt{2x^6 + 3x^4 + 6x^2} dx = \int_0^1 (x^5 + x^3 + x) \sqrt{2x^6 + 3x^4 + 6x^2} dx$$

Let $u = 2x^6 + 3x^4 + 6x^2$. Then $du = (12x^5 + 12x^3 + 12x) dx = 12(x^5 + x^3 + x) dx$. The limits change to $u(0) = 0$ and $u(1) = 2 + 3 + 6 = 11$:

$$I = \frac{1}{12} \int_0^{11} u^{1/2} du = \frac{1}{12} \left[\frac{2}{3} u^{3/2} \right]_0^{11} = \frac{1}{18} \cdot 11\sqrt{11} = \frac{11\sqrt{11}}{18}$$

1.3 8 July 2026

Medium Integral

$$I = \int_{1/4}^4 \frac{1}{x + 3^{\ln(x)}} dx$$

Solution: Recognize that $3^{\ln(x)} = e^{\ln(x) \ln(3)} = x^{\ln(3)}$. The integral rewrites as:

$$I = \int_{1/4}^4 \frac{1}{x + x^{\ln(3)}} dx$$

Multiply the numerator and denominator by $x^{-\ln(3)}$:

$$I = \int_{1/4}^4 \frac{x^{-\ln(3)}}{x^{1-\ln(3)} + 1} dx$$

Let $u = x^{1-\ln(3)} + 1$. The differential is $du = (1 - \ln(3))x^{-\ln(3)} dx$. This transforms the integral into a standard logarithmic form:

$$I = \frac{1}{1 - \ln(3)} \left[\ln(x^{1-\ln(3)} + 1) \right]_{1/4}^4$$

Evaluate at the bounds: At $x = 4$: $\ln(4^{1-\ln(3)} + 1)$

At $x = 1/4$: $\ln\left((1/4)^{1-\ln(3)} + 1\right) = \ln\left(4^{\ln(3)-1} + 1\right) = \ln\left(\frac{1+4^{1-\ln(3)}}{4^{1-\ln(3)}}\right)$ Subtracting the lower bound from the upper bound using logarithm properties:

$$\ln\left(4^{1-\ln(3)} + 1\right) - \ln\left(\frac{1+4^{1-\ln(3)}}{4^{1-\ln(3)}}\right) = \ln\left(4^{1-\ln(3)}\right) = (1 - \ln(3)) \ln(4)$$

Multiply this by the constant factor in front:

$$I = \frac{1}{1 - \ln(3)} \cdot (1 - \ln(3)) \ln(4) = \ln(4) = 2 \ln(2)$$

1.4 7 July 2026

Medium Integral

$$I = \int_{-3/2}^{7/2} \frac{\sin\left(\frac{\pi x}{2}\right)}{1 + e^{x^3 - 3x^2 + 2x}} dx$$

Solution: Let the bounds be $a = -3/2$ and $b = 7/2$. The midpoint of the interval is $c = \frac{a+b}{2} = 1$. We use the substitution property $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, where $a+b = 2$. Let $x \rightarrow 2-x$. We analyze how the components of the integrand change. For the exponent polynomial $P(x) = x^3 - 3x^2 + 2x = x(x-1)(x-2)$:

$$P(2-x) = (2-x)(2-x-1)(2-x-2) = (2-x)(1-x)(-x) = -x(x-1)(x-2) = -P(x)$$

For the trigonometric function:

$$\sin\left(\frac{\pi(2-x)}{2}\right) = \sin\left(\pi - \frac{\pi x}{2}\right) = \sin\left(\frac{\pi x}{2}\right)$$

The transformed integral is:

$$I = \int_{-3/2}^{7/2} \frac{\sin\left(\frac{\pi x}{2}\right)}{1 + e^{-P(x)}} dx = \int_{-3/2}^{7/2} \frac{e^{P(x)} \sin\left(\frac{\pi x}{2}\right)}{e^{P(x)} + 1} dx$$

Adding the original integral to this transformed version:

$$2I = \int_{-3/2}^{7/2} \frac{(1 + e^{P(x)}) \sin\left(\frac{\pi x}{2}\right)}{1 + e^{P(x)}} dx = \int_{-3/2}^{7/2} \sin\left(\frac{\pi x}{2}\right) dx$$

Evaluating this simple integral:

$$\begin{aligned} 2I &= \left[-\frac{2}{\pi} \cos\left(\frac{\pi x}{2}\right) \right]_{-3/2}^{7/2} = -\frac{2}{\pi} \left(\cos\left(\frac{7\pi}{4}\right) - \cos\left(-\frac{3\pi}{4}\right) \right) \\ 2I &= -\frac{2}{\pi} \left(\frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) \right) = -\frac{2}{\pi}(\sqrt{2}) \implies I = -\frac{\sqrt{2}}{\pi} \end{aligned}$$

1.5 6 July 2026

Medium Integral

$$I = \int_0^{\pi/2} \left(\frac{\int_0^x \ln(2 \sin(t)) dt}{\int_0^{2x} \ln(2 \sin(t)) dt} + \frac{\int_0^{\pi/4} \ln(\cot(t)) dt}{(4x - \pi) \ln(2)} \right) dx$$

Solution: Let $L_1(x) = \int_0^x \ln(2 \sin t) dt$ and $L_2(x) = \int_0^x \ln(2 \cos t) dt$. Using the double angle identity, their sum is:

$$L_1(x) + L_2(x) = \int_0^x \ln(4 \sin t \cos t) dt = \int_0^x \ln(2 \sin 2t) dt = \frac{1}{2} \int_0^{2x} \ln(2 \sin u) du$$

Let $G(x) = \int_0^{2x} \ln(2 \sin t) dt$. Thus, $G(x) = 2(L_1(x) + L_2(x))$. The first term of our integrand is $f(x) = \frac{L_1(x)}{G(x)}$. Apply the reflection $x \rightarrow \frac{\pi}{2} - x$:

$$L_1(\pi/2 - x) = \int_0^{\pi/2-x} \ln(2 \sin t) dt = -L_2(x) \quad \text{and} \quad G(\pi/2 - x) = -G(x)$$

Thus, $f(\frac{\pi}{2} - x) = \frac{-L_2(x)}{-G(x)} = \frac{L_2(x)}{G(x)}$. Notice that $f(x) + f(\frac{\pi}{2} - x) = \frac{L_1(x) + L_2(x)}{G(x)} = \frac{1}{2}$. Now analyze the second term. The numerator is Catalan's Constant, $K = \int_0^{\pi/4} \ln(\cot t) dt$. Let $S(x) = \frac{K}{(4x - \pi) \ln 2}$. Reflecting $x \rightarrow \frac{\pi}{2} - x$ yields:

$$S(\pi/2 - x) = \frac{K}{(4(\pi/2 - x) - \pi) \ln 2} = \frac{K}{(\pi - 4x) \ln 2} = -S(x)$$

Thus, $S(x) + S(\frac{\pi}{2} - x) = 0$. The apparent pole at $x = \pi/4$ perfectly cancels between the two terms. For the overall integrand $H(x) = f(x) + S(x)$, we have $H(x) + H(\frac{\pi}{2} - x) = \frac{1}{2}$.

$$2I = \int_0^{\pi/2} \left(H(x) + H\left(\frac{\pi}{2} - x\right) \right) dx = \int_0^{\pi/2} \frac{1}{2} dx = \frac{\pi}{4} \implies I = \frac{\pi}{8}$$

Hard Integrals

2.1 10 July 2026

Hard Integral

$$I = \int_0^1 \tan^{-1}(x) \text{Li}_2(x) dx$$

Solution: We use integration by parts. Let $u = \text{Li}_2(x)$ and $dv = \tan^{-1}(x) dx$. The derivatives and antiderivatives are:

$$du = -\frac{\ln(1-x)}{x} dx$$

$$v = x \tan^{-1}(x) - \frac{1}{2} \ln(1+x^2)$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$I = \left[\text{Li}_2(x) \left(x \tan^{-1}(x) - \frac{1}{2} \ln(1+x^2) \right) \right]_0^1 + \int_0^1 \frac{\ln(1-x)}{x} \left(x \tan^{-1}(x) - \frac{1}{2} \ln(1+x^2) \right) dx$$

Evaluate the boundary term. At $x = 0$, it evaluates to 0. At $x = 1$, using $\text{Li}_2(1) = \frac{\pi^2}{6}$:

$$\text{Boundary Term} = \frac{\pi^2}{6} \left(\frac{\pi}{4} - \frac{\ln(2)}{2} \right) = \frac{\pi^3}{24} - \frac{\pi^2 \ln(2)}{12}$$

The remaining integral can be challenging to express in simple elementary forms, leading us back to a series representation. We express the overall integral utilizing the Maclaurin series for $\tan^{-1}(x)$:

$$\int_0^1 \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} \text{Li}_2(x) dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \int_0^1 x^{2n+1} \text{Li}_2(x) dx$$

Using the known integral reduction $\int_0^1 x^k \text{Li}_2(x) dx = \frac{\zeta(2)}{k+1} - \frac{H_{k+1}}{(k+1)^2}$ where H_k is the k -th harmonic number:

$$I = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \left(\frac{\pi^2}{6(2n+2)} - \frac{H_{2n+2}}{(2n+2)^2} \right)$$

Separating the summation into two distinct series:

$$I = \frac{\pi^2}{12} \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2n+1} - \frac{1}{2n+2} \right) - \sum_{n=0}^{\infty} \frac{(-1)^n H_{2n+2}}{(2n+1)(2n+2)^2}$$

The first sum collapses and evaluates to $\frac{\pi}{4} - \frac{\ln(2)}{2}$, perfectly mirroring the boundary term derived from our initial integration by parts! Thus, the exact analytical representation of this integral in terms of an alternating harmonic Euler sum is:

$$I = \frac{\pi^3}{24} - \frac{\pi^2 \ln(2)}{12} - \sum_{n=0}^{\infty} \frac{(-1)^n H_{2n+2}}{(2n+1)(2n+2)^2}$$

2.2 9 July 2026

Hard Integral

$$I = \int_0^1 \ln(\{\ln(x)\}) dx$$

Solution: Let $u = -\ln(x)$, meaning $x = e^{-u}$ and $dx = -e^{-u} du$. The fractional part $\{\ln(x)\}$ becomes $\{-u\}$. The limits transform to 0 to ∞ :

$$I = \int_0^\infty \ln(\{-u\}) e^{-u} du$$

Split the integral into unit intervals where $u \in [n, n+1]$. In this interval, $\{-u\} = (n+1) - u$.

$$I = \sum_{n=0}^{\infty} \int_n^{n+1} \ln(n+1-u) e^{-u} du$$

Let $v = n+1-u \implies du = -dv$. The limits shift from 1 down to 0:

$$I = \sum_{n=0}^{\infty} \int_0^1 \ln(v) e^{-(n+1-v)} dv = \left(\sum_{n=0}^{\infty} e^{-(n+1)} \right) \int_0^1 e^v \ln(v) dv$$

The geometric series evaluates to $\frac{e^{-1}}{1-e^{-1}} = \frac{1}{e-1}$. For the remaining integral, we use integration by parts, acknowledging the improper limit at $v \rightarrow 0^+$:

$$\int_0^1 e^v \ln(v) dv = [e^v \ln(v)]_0^1 - \int_0^1 \frac{e^v}{v} dv$$

To handle the divergence correctly, we employ the Exponential Integral definition $\text{Ei}(1) = \gamma + \int_0^1 \frac{e^v - 1}{v} dv$. Evaluating the boundary alongside the divergent piece cancels the singularity:

$$\int_0^1 e^v \ln(v) dv = \gamma - \text{Ei}(1)$$

Combining these results yields the closed-form:

$$I = \frac{\gamma - \text{Ei}(1)}{e-1}$$

2.3 8 July 2026

Hard Integral

$$I = \int_0^{\pi/4} \frac{\sin(x + \pi/4)}{\cos^3(x)} \{\cot(x)\} dx$$

Solution: Expand $\sin(x + \pi/4) = \frac{1}{\sqrt{2}}(\sin(x) + \cos(x))$. Substitute this into the integral:

$$I = \frac{1}{\sqrt{2}} \int_0^{\pi/4} (\tan(x) \sec^2(x) + \sec^2(x)) \{\cot(x)\} dx$$

Let $u = \cot(x)$, then $du = -\csc^2(x) dx = -(1+u^2) dx$. Notice $\sec^2(x) = 1 + \frac{1}{u^2}$. After algebraic manipulation, the integrand simplifies drastically. The boundaries change from ∞ to 1:

$$I = \frac{1}{\sqrt{2}} \int_1^\infty \frac{u+1}{u^3} \{u\} du = \frac{1}{\sqrt{2}} \sum_{n=1}^{\infty} \int_n^{n+1} \frac{u+1}{u^3} (u-n) du$$

Evaluate the integral $F(u) = \int \frac{u^2 + (1-n)u - n}{u^3} du = \ln(u) - \frac{1-n}{u} + \frac{n}{2u^2}$. Evaluating across $[n, n+1]$ yields the summand $F(n+1) - F(n)$:

$$\ln\left(\frac{n+1}{n}\right) - \frac{3}{2(n+1)} - \frac{1}{2(n+1)^2} + \frac{1}{2n}$$

Summing from $n = 1$ to N : 1. The logarithmic term telescopes to $\ln(N+1)$. 2. The term $\frac{1}{2n} - \frac{1}{2(n+1)}$ telescopes to $\frac{1}{2} - \frac{1}{2(N+1)}$. 3. The remaining terms format into convergent series tied to the Euler-Mascheroni constant γ and the Basel problem:

$$-\sum_{k=2}^{N+1} \frac{1}{k} - \frac{1}{2} \sum_{k=2}^{N+1} \frac{1}{k^2} \approx -(\ln(N+1) + \gamma - 1) - \frac{1}{2} \left(\frac{\pi^2}{6} - 1 \right)$$

Taking the limit as $N \rightarrow \infty$, the $\ln(N+1)$ cancels perfectly:

$$\text{Sum} = \frac{1}{2} - \gamma + 1 - \frac{\pi^2}{12} + \frac{1}{2} = 2 - \gamma - \frac{\pi^2}{12}$$

Multiplying by the leading constant gives the exact analytical answer:

$$I = \sqrt{2} - \frac{\gamma}{\sqrt{2}} - \frac{\pi^2}{12\sqrt{2}}$$

2.4 7 July 2026

Hard Integral

$$I = \int_0^1 \frac{\ln\left(\frac{2+\sqrt{3}x}{2-\sqrt{3}x}\right)}{x\sqrt{1-x^2}} dx$$

Solution: Let $x = \sin(\theta)$, $dx = \cos(\theta) d\theta$. The denominator $x\sqrt{1-x^2}$ becomes $\sin(\theta) \cos(\theta)$. The interval maps to $0 \rightarrow \pi/2$:

$$I = \int_0^{\pi/2} \frac{\ln\left(\frac{2+\sqrt{3}\sin\theta}{2-\sqrt{3}\sin\theta}\right)}{\sin\theta} d\theta$$

To evaluate this, introduce a parameter a and define a function using Leibniz's rule (Feynman's Trick):

$$f(a) = \int_0^{\pi/2} \frac{\ln\left(\frac{1+a\sin\theta}{1-a\sin\theta}\right)}{\sin\theta} d\theta$$

Differentiate with respect to a :

$$f'(a) = \int_0^{\pi/2} \frac{1}{\sin\theta} \left(\frac{\sin\theta}{1+a\sin\theta} + \frac{\sin\theta}{1-a\sin\theta} \right) d\theta = \int_0^{\pi/2} \frac{2}{1-a^2\sin^2\theta} d\theta$$

This is a standard trigonometric integral that resolves to $\frac{\pi}{\sqrt{1-a^2}}$. Integrating back with respect to a from 0 to A :

$$f(A) = \int_0^A \frac{\pi}{\sqrt{1-a^2}} da = \pi \arcsin(A)$$

In our original integral, divide the numerator argument by 2 to see $a = \frac{\sqrt{3}}{2}$:

$$I = f\left(\frac{\sqrt{3}}{2}\right) = \pi \arcsin\left(\frac{\sqrt{3}}{2}\right) = \pi\left(\frac{\pi}{3}\right) = \frac{\pi^2}{3}$$

2.5 6 July 2026

Hard Integral

$$I = \int_0^\infty \frac{\{x^2\}}{x^2} dx$$

Solution: Let $u = x^2$, which gives $dx = \frac{du}{2\sqrt{u}}$. The integral transforms into:

$$I = \frac{1}{2} \int_0^\infty \frac{\{u\}}{u^{3/2}} du = \frac{1}{2} \sum_{n=0}^\infty \int_n^{n+1} \frac{u-n}{u^{3/2}} du$$

Compute the generic term for $n \geq 1$:

$$\int_n^{n+1} (u^{-1/2} - nu^{-3/2}) du = \left[2u^{1/2} + 2nu^{-1/2} \right]_n^{n+1}$$

Substituting the bounds simplifies algebraically to:

$$4\sqrt{n+1} - 4\sqrt{n} - \frac{2}{\sqrt{n+1}}$$

Summing this from $n = 1$ to N , the first two terms heavily telescope:

$$4\sqrt{N+1} - 4 - 2 \sum_{k=2}^{N+1} \frac{1}{\sqrt{k}}$$

Using the generalized Euler-Maclaurin expansion for the Riemann Zeta Function evaluated at $s = 1/2$, we know that $\sum_{k=1}^M \frac{1}{\sqrt{k}} \approx 2\sqrt{M} + \zeta(1/2)$. Substituting this asymptotic equivalence into our sum:

$$\lim_{N \rightarrow \infty} \left(4\sqrt{N+1} - 4 - 2 \left(2\sqrt{N+1} + \zeta(1/2) - 1 \right) \right) = -2 - 2\zeta(1/2)$$

Adding the $n = 0$ integral block, which simply evaluates to 2:

$$\text{Total Limit} = 2 - 2 - 2\zeta(1/2) = -2\zeta(1/2)$$

Multiplying by the global $1/2$ factor gives the astoundingly clean result:

$$I = -\zeta\left(\frac{1}{2}\right)$$